

Assignment - 3

Q.1 Show that the function $f(z) = xy + iy$ is nowhere analytic.

Q.2 Where does the function $f(z) = \frac{z^3 - 1}{z^3 + z}$ fail to be analytic?

Q.3 Show that $v(x, y) = 2y - 3x^2y + y^3$ is harmonic.

Q.4 Show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic.

Q.5 Determine the singular points of the function
$$f(z) = \frac{z^3 + 4}{(z^2 - 3)(z^2 + 1)}$$

Q.6 Show that the function $f(z) = 2xy + i(x^2 - y^2)$ is nowhere analytic.

Q.7 Evaluate the definite integral

$$\int_0^{\pi/6} e^{2it} dt$$

Q.8 Evaluate $\int_C \frac{z+2}{z} dz$, where $C: z = 2e^{i\theta}; 0 \leq \theta \leq \pi$

Q.9 Let C be the unit circle $|z|=1$. Apply the Cauchy-Goursat theorem to show that

$$\int_C \frac{z^2}{z-3} dz = 0$$

Q.10 Evaluate the definite integral

$$\int_1^2 \left(\frac{1}{t} - t \right)^2 dt$$

Q.11 Evaluate $\int_C \bar{z} dz$, where $C: z = 2e^{i\theta}$; $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

Q.12 Let C be the unit circle $|z|=1$. Apply the Cauchy-Goursat theorem to show that

$$\int_C z e^{-z} dz = 0$$

Q.13 Evaluate using Cauchy integral formula

$$\int_C \frac{1}{z^2+4} dz, \text{ where } C \text{ is the positively oriented circle } |z-i|=2$$

Q.14 Evaluate $\int_C \frac{e^{2z}}{z^4} dz$, where C is the positively oriented circle $|z|=1$

Q.15 Using Cauchy-Integral formula to evaluate

$$\int_C \frac{z}{(9-z^2)(z+i)} dz, \text{ where } C \text{ is the positively oriented circle } |z|=2.$$

Q.16 Evaluate $\int_C \frac{z^2+2}{(z-1)^3} dz$, where C is the positively oriented circle $|z|=2$.

Q.17 Define Entire function. State the Liouville's theorem.

Q.18 State the fundamental theorem of algebra. Find the zeros of the polynomial $f(z) = (z^3+z)(z-1)$

Q.19 Construct the Maclaurin series for the function $f(z) = e^z$

Q.20 Obtain the ~~the~~ Taylor series for the function $f(z) = z^3$ about $z = 1$.

Q.21 Construct the Maclaurin series for the function $f(z) = \frac{1}{1-z}$ when $|z| < 1$

Q.22 Construct the Taylor series for the function $f(z) = z^3$ about $z = -1$

Q.23 Find the Laurent series representation of $f(z) = \frac{1}{1+z}$ when $0 < |z| < 1$.

Q.24 Find the Laurent series representation of $f(z) = \frac{1}{z^2(1-z)}$, when $0 < |z| < 1$.

Q.25 Determine the isolated singularities of the function $f(z) = \frac{e^z - 1}{z(z-2)}$. Explain the types of those singular points.

Q.26 Determine the isolated singularities of the function $f(z) = \frac{\sin z}{z^3}$. Also explain their types.

Q.27 Find out the residue of the function $f(z) = \frac{z^2 + 5z}{z-5}$ at $z = 5$.

Q.28 Determine the poles & their order for the function $f(z) = \frac{5z^3 + 1}{(z-5)^2}$. Find the residue at the poles.

Q.29 Find out the residue of the function $f(z) = \frac{z^3}{z-2}$ at $z = 2$.

Q.30 Determine the poles and their order for the function $f(z) = \frac{2z^2 + 3z}{(z-2)^2}$. Find the residue at the poles.